Machine Learning 2026 • Tutorial 1

Mathematical Readiness and A0 Clinic **SOLUTIONS**

07 August 2026 • B2, 08:30–09:50 • Jagdish/Harsha Patel Learning Theatres

Everything here is plain mathematics. Nothing needs machine learning today — and every piece returns later wearing a machine-learning name. **No calculator is needed:** every value you cannot do in your head is given to you. Work in the boxes; bring one unresolved question to Tutorial 2.

Station 1 Shapes: when does a product exist?

A product $\mathbf{AB}$ with $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{B} \in \mathbb{R}^{p\times q}$ exists only if $n = p$, and then $\mathbf{AB} \in \mathbb{R}^{m\times q}$. *Inner dimensions agree; outer dimensions survive.* A column vector $\mathbf{v} \in \mathbb{R}^d$ is a $d \times 1$ matrix, so $\mathbf{u}^\top\mathbf{v}$ is a scalar while $\mathbf{uv}^\top$ is $d \times d$.

1.1 For $\mathbf{A} \in \mathbb{R}^{3\times 2}$ and $\mathbf{B} \in \mathbb{R}^{2\times 4}$: which of $\mathbf{AB}$, $\mathbf{BA}$ exists, and with what shape?

Answer. $\mathbf{AB}$ exists (inner $2 = 2$), shape 3×4. $\mathbf{BA}$ does not: inner $4 \neq 3$.

1.2 Let $\mathbf{X} \in \mathbb{R}^{n\times d}$, $\theta \in \mathbb{R}^d$, $\mathbf{y} \in \mathbb{R}^n$. Fill in the dimensions.

Expression	Dimension	Meaning
$\mathbf{X}\theta$	$n \times 1$	one prediction per observation
$\mathbf{y} - \mathbf{X}\theta$	$n \times 1$	residual (error) vector
$\mathbf{X}^\top\mathbf{y}$	$d \times 1$	feature–target aggregate
$\mathbf{X}^\top\mathbf{X}$	$d \times d$	feature–feature aggregate

Which of $\mathbf{Xy}$, $\mathbf{X}^\top\mathbf{y}$, $\theta^\top\mathbf{X}^\top\mathbf{y}$ is invalid, and why?

Answer. $\mathbf{Xy}$ is invalid: inner $d \neq n$ in general. The other two are fine; $\theta^\top\mathbf{X}^\top\mathbf{y}$ is a scalar.

1.3 Let $\mathbf{x} \in \mathbb{R}^4$, $\mathbf{W}_1 \in \mathbb{R}^{6\times 4}$, $\mathbf{b}_1 \in \mathbb{R}^6$, $\mathbf{W}_2 \in \mathbb{R}^{2\times 6}$, $\mathbf{b}_2 \in \mathbb{R}^2$. Give the shape of every intermediate object in $\mathbf{z} = \mathbf{W}_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$. Could you multiply in the other order?

Answer. $\mathbf{W}_1\mathbf{x} : 6 \times 1$; $+\mathbf{b}_1$ keeps 6×1; $\mathbf{W}_2(\cdot) : 2 \times 1$; $+\mathbf{b}_2$ keeps 2×1. So $\mathbf{z} \in \mathbb{R}^2$. Reverse order fails: $\mathbf{W}_1\mathbf{W}_2$ has inner $4 \neq 2$.

Later in this course: *this chain is the* forward pass of a two-layer network *— four inputs, six hidden units, two outputs.*

Station 2 Two functions, studied as functions

2.1 Let $\sigma(z) = \dfrac{1}{1 + e^{-z}}$. Given $e^{-2} \approx 0.135$ and $e^2 \approx 7.39$, compute $\sigma(0)$, $\sigma(2)$, $\sigma(-2)$; state both limits as $z \to \pm\infty$; and check $\sigma(-z) = 1 - \sigma(z)$ on your numbers.

Answer. $\sigma(0) = 0.5$; $\sigma(2) = 1/1.135 = 0.881$; $\sigma(-2) = 1/8.389 = 0.119$. Limits 1 and 0. Check: $0.881 + 0.119 = 1.000$. (Also $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, so the slope at 0 is $1/4$ — steepest at the midpoint.)

Later in this course: *σ squeezes $\mathbb{R}$ into $(0, 1)$ and returns as the function that turns a score into a probability.*

2.2 Let $m(z) = \max(0, z)$. Sketch it (axes below); compute $m(-3)$, $m(0)$, $m(2)$; give $m'(z)$ for $z > 0$ and $z < 0$; then say what $m'(0)$ is — be honest.

Answer. $m(-3) = 0$, $m(0) = 0$, $m(2) = 2$. $m'(z) = 1$ for $z > 0$, 0 for $z < 0$. $m'(0)$ *does not exist*: the slope jumps from 0 to 1.

Later in this course: *this is ReLU, the most used activation in deep networks — and that corner will matter.*

Station 3 Finding extrema, and checking them

The test. If f is twice differentiable and $f'(u^\star) = 0$, then $f''(u^\star) > 0$ gives a local minimum, $f''(u^\star) < 0$ a local maximum, and $f''(u^\star) = 0$ is inconclusive.

3.1 For $h(w) = (w-3)^2 + 2$: find the critical point, apply the test, and state the minimum value.

> **Answer.** $h'(w) = 2(w-3) = 0 \Rightarrow w = 3$; $h'' = 2 > 0$, a minimum; $h(3) = 2$.

3.2 For $g(u) = u^4 - 2u^2$: find *every* critical point, classify each with the test, give g at each, and say how many *global* minima there are.

> **Answer.** $g'(u) = 4u(u-1)(u+1) = 0$ at $u = -1, 0, 1$. $g''(u) = 12u^2 - 4$: at $u = \pm 1$, $g'' = 8 > 0$ (two local minima); at $u = 0$, $g'' = -4 < 0$ (a local maximum). $g(\pm 1) = -1$, $g(0) = 0$. Both valleys reach -1, so there are **two** global minima.

Later in this course: *"one valley" is called* convexity*. Objectives with many valleys are why optimization in machine learning is genuinely hard. The derivations come later — the test you can already run.*

Station 4 A Bayes calculation, start to finish

Floods occur on 10% of monsoon days. Rainfall behaves differently on the two kinds of day:

$$\text{flood: } x \sim \mathcal{N}(120, 30^2) \text{ mm}, \qquad \text{no flood: } x \sim \mathcal{N}(40, 20^2) \text{ mm}, \qquad p(x) = \frac{c}{s}\, e^{-(x-m)^2/(2s^2)}.$$

Today it rained $x = 90$ mm. *You will never need the constant c* — watch it cancel. Values you may need, all given: $e^{2.625} \approx 13.8$, $e^{-1.5} \approx 0.223$, $e^{8} \approx 2981$.

4.1 Write the exponent $-\dfrac{(x-m)^2}{2s^2}$ at $x = 90$ under each hypothesis. (Both are clean fractions.)

> **Answer.** Flood: $-\dfrac{(90-120)^2}{2(30)^2} = -\dfrac{900}{1800} = \mathbf{-0.5}$. No flood: $-\dfrac{(90-40)^2}{2(20)^2} = -\dfrac{2500}{800} = \mathbf{-3.125}$. Note 90 is one standard deviation from the flood mean, 2.5 from the other.

4.2 Form the ratio $\dfrac{p(90 \mid \text{flood})}{p(90 \mid \text{no flood})}$. Show that c cancels, that the standard deviations leave $\frac{20}{30}$, and that the exponents combine into a single power of e. Evaluate it.

> **Answer.** $\dfrac{(c/30)e^{-0.5}}{(c/20)e^{-3.125}} = \dfrac{20}{30}\, e^{-0.5+3.125} = \frac{2}{3}\, e^{2.625} \approx \frac{2}{3}(13.8) = \mathbf{9.2}$. So 90 mm is about nine times more typical of a flood day.

4.3 The prior odds of flood against no flood are $0.10 : 0.90$, i.e. $1 : 9$. Multiply prior odds by the ratio from 4.2 to get the posterior odds, then convert odds to a probability using $p = \dfrac{\text{odds}}{1 + \text{odds}}$.

> **Answer.** Posterior odds $= \dfrac{1}{9} \times 9.2 = 1.02$. Then $p(\text{flood} \mid 90) = \dfrac{1.02}{1 + 1.02} = \dfrac{1.02}{2.02} \approx \mathbf{0.51}$. The evidence says flood nine-to-one; the base rate says no-flood nine-to-one; they cancel, and the day is a coin flip.

4.4 (homework) Repeat 4.1–4.3 at $x = 60$ and $x = 120$ mm.

> **Answer.** At 60: exponents -2 and -0.5, ratio $\frac{2}{3}e^{-1.5} \approx 0.15$, odds $0.15/9 = 0.017$, $p \approx 0.016$. At 120: exponents 0 and -8, ratio $\frac{2}{3}e^{8} \approx 1987$, odds ≈ 221, $p \approx 0.996$. Far from the overlap the base rate wins; inside it the evidence wins.

Later in this course: *this is exactly how a* generative *model predicts — model how each kind of day produces data, then invert with Bayes' rule. The odds form you used is the same rule, rearranged so the constants never appear.*

Where today points. A0 (Mathematical Readiness) is open on the portal, due **14 August, 23:59 IST**. Shapes → vectors, matrices and dimension checks. Functions → reading and evaluating functions. Extrema → derivatives and optimization. Bayes → probability and conditioning.

Response standard. For every expression you write: name the object type, state its dimension, and state the assumption behind the next step.